RF-U diagrams for calibration of the material models

Alexandros Stathas Matthias Neuner

April 2026

1 Introduction

In order to facilitate calibration of the different modeling approaches, we provide the reaction force displacement curves (RF-U) for a box specimen. The specimen has dimensions 200 mm x 1 mm x 1 mm (see Figure 1). We discretize the model with a single column of finite elements using the GCDP model (GCPS8R) in plane stress, and apply a nodal displacement on the right side of the model keeping the left side fixed. We evaluate the fracture energy from the tension experiment as the area under the metelastic branch of the RF-U curve by numerical integration. We produce also the RF-U curves for a specimen under compression for the same given fracture energy G_F so the different modeling approaches can be calibrated in tension - compression.

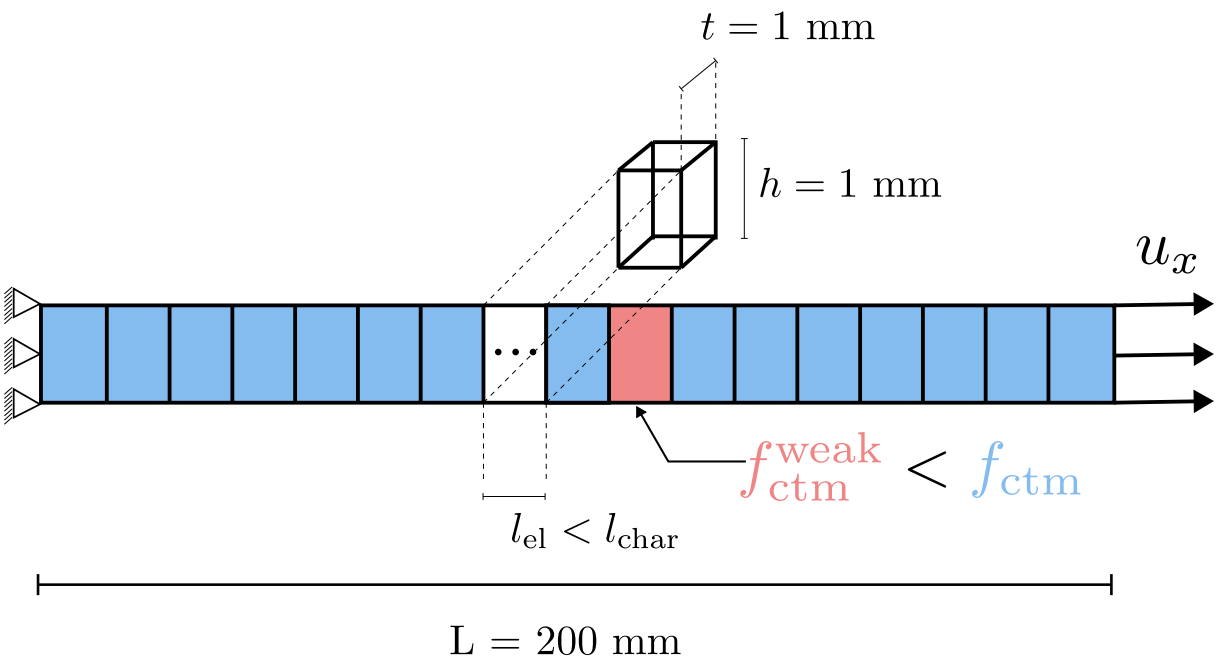


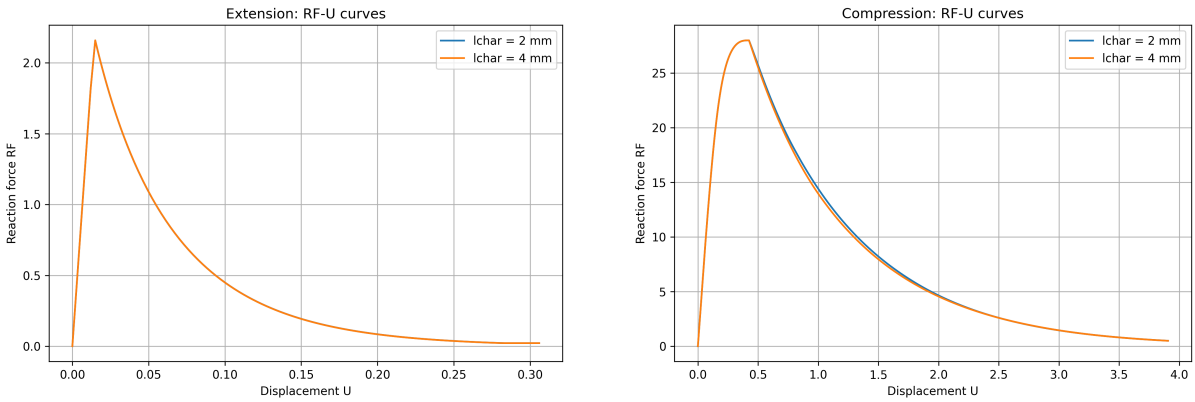
Figure 1: Geometry of the model bar. The Model is discretized with plain stress elements so that $l_{\text{el}} < l_{\text{char}}$. For $l_{\text{char}} = 2$ mm, $l_{\text{el}} = \frac{200}{260}$ mm. For $l_{\text{char}} = 4$ mm, $l_{\text{el}} = \frac{200}{130}$ mm. The element in the middle of the model (red) has a weakened mean tensile strength, $f_{\text{ctm}}^{\text{weak}} = 0.99 f_{\text{ctm}}$, in order to promote localization. The displacement field on the left of the bar is set zero and an extensive (compressive) horizontal displacement is applied on the right.

1.1 Material properties

We perform the calibration of the characteristic length and softening modulo for the GCDP model for the given fracture energy of concrete grade C20/25, $G_F = 133$ N/mm. We choose two different values of the characteristic length $l_{\text{char}} = 2$ mm, $l_{\text{char}} = 4$ mm for which -based on the value of the fracture energy- we obtain the corresponding softening moduli $\varepsilon_{\text{sf}} = 0.01205$ and $\varepsilon_{\text{sf}} = 0.00675$. Keeping the two parameter sets allows us to investigate different mesh discretizations in particular larger values of l_{char} allow us to increase the mesh size reducing computational demands if necessary. We emphasize that all analyses regarding the pryout tests have been conducted with the smallest characteristic length $l_{\text{char}} = 2$ mm.

2 RF-U curves results

The results of analysis of the bar model are given in Figure 2 for extension and compression. The .csv data for the creation of the plots ar in the child folder ./RF-U curves.



(a) (b)

Figure 2: RF-U curves for a specimen in a) tension and b) compression for the same fracture energy. Dimensions of the specimen: Length 200 mm, section height 1 mm and section width 1 mm. RF-U curves are given for concrete C20/25 with fracture energy $G_F = 133$ N/mm.